Early Heart Stroke Warning Wearable

BM2210 – Biomedical Device Designing

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Team Vitasync

VitaSync

The Need

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Need Statement

A way to continuously monitor the onset of anhidrosis (sweat cessation) and critical core body temperature shifts in individuals exposed to high environmental heat or prolonged physical exertion, to provide actionable early warnings of heat stroke before physiological collapse occurs.

Why Today’s Options Fall Short

WBGT monitors
Measure the environment, not the person

Smart Watches
Miss the individual's heat state

Core-temp pills
Invasive and expensive to scale

Our Solution

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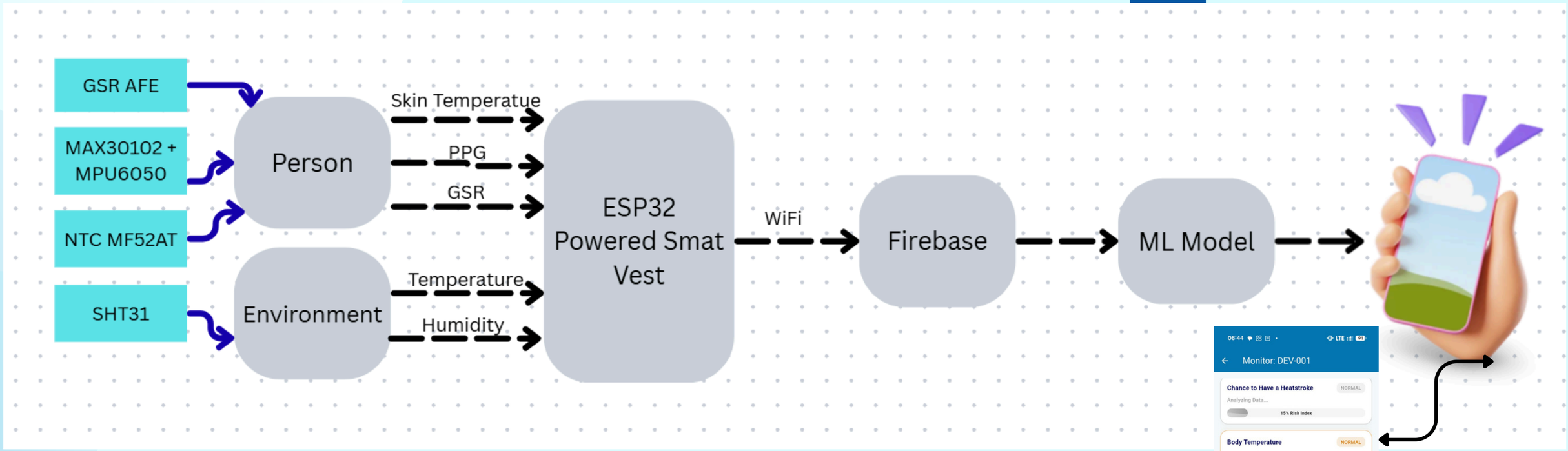
- **A low-cost, non-invasive wearable** ecosystem paired with a **smart mobile application** that addresses the critical gap in heat stroke prevention.

- The system continuously tracks physiological and environmental parameters including **galvanic skin response, environmental heat index, photoplethysmography (PPG), and skin temperature** to flag the earliest precursors of heat stress such as the onset of anhidrosis.

- Finally the risk level predicted by the **machine learning** model is written back to Firebase for the app.

Our Solution

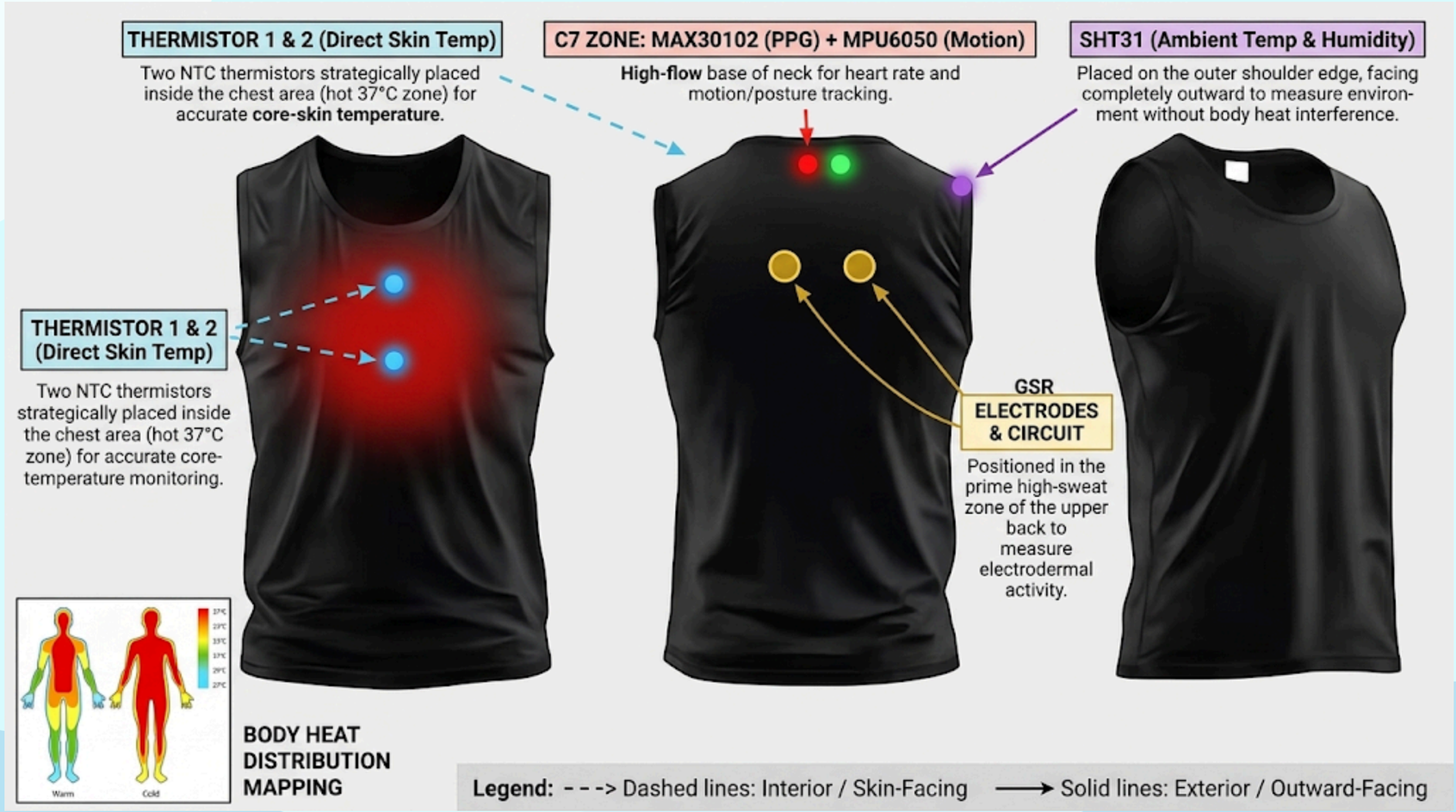
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Our Solution

Sensor Placing on Smart Vest



Stake Holders

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- **Primary Users:** Outdoor laborers (construction, agricultural), military personnel, athletes/cricketers in high-heat areas, and elderly individuals.



- **Decision Makers / Buyers:** OHS managers, company executives, military procurement officers, and elderly care administrators.



- **Secondary Stakeholders:** Insurance companies (aiming to reduce heat-related injury claims) and labor unions.



Regulations

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Chosen Strategy (General Wellness):

- Pursuing the FDA general wellness exemption to position the system as a monitoring and evaluation tool for occupational and athletic settings.
- Bypasses extensive screening and diagnosis, prevention, or mitigation claims.

Avoided Path (Medical Device Claim):

- Avoiding an FDA Class II device classification to prevent long review cycles, full ISO 13485 requirements, clinical trials, and 6 to 18 months of delay in revenue generation.

Local Compliance Requirements:

- Mandatory registration with the NMRA (Sri Lanka) taking at least 6 months through a local marketing authority holder.
- Requires submitting a sample import license and a full product dossier.

Business Model & Strategy

- B2B hardware-focused model involving smart vest sales/leases, with a free accompanying manager fleet dashboard and mobile application.

- **Screening Results:**
  - Occupational Safety = Go (+2)
  - Elite Athletics = Revisit (0)
  - Consumer = No-go (-2).

- **IP Protection:** Utility and design patents filed via NIPO/PCT route with a freedom-to-operation check.

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Validation Plan

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- **Benchtop Testing (Cross-talk Test):**
  - *Environment:* Controlled thermal chamber sweep from 25 °C to 45 °C.
  - *Pass Criteria:* Temperature accuracy within ±0.2 °C and total filtering/rejection of false GSR (sweat) alarms.

- **Active Exertion Testing (Motion & Light Test):**
  - *Environment:* High-movement physical activity (running and jumping) under direct sunlight, benchmarked against a chest strap monitor.
  - *Pass Criteria:* Heart rate measurement accuracy within ±4 bpm.

- **Field Endurance Testing (Heat Exertion Protocol):**
  - *Environment:* Continuous 4 hour real world heat exertion field deployment.
  - *Pass Criteria:* Trigger safety alerts in under 2 seconds with zero skin irritation for the user.

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Q & A